MIL-H-80055B
20 March 1981
Superseding
MIL-H-80055A
8 May 1969

MILITARY SPECIFICATION

HEATERS, INDUCTION, HIGH FREQUENCY; MOTOR GENERATOR TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high frequency induction heaters consisting of a high frequency motor generator (MG) set, a starter, a control station, and a heating station.

* 1.2 Classification. The induction heaters covered by this specification are classified in sizes from 7 1/2 through 50 kilowatts (KW) and of the following types. The type and size required shall be as specified (see 3.6 and 6.2.1).

Type I - Generator output frequency, 3 kilohertz (KHz)
Type II - Generator output frequency, 10 KHz

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3424

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STANDARDS

FEDERAL

* FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

* MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents, form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

* U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

Title 29 Code of Federal Regulations, Chapter XVII, Part 1910, and Amendments	Occupational Safety and Health Administration Standards

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

* AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-380 - Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103).

NATIONAL ELECTRICAL. MANUFACTURER'S ASSOCIATION (NEMA)

MG-1 - Motors and Generators
ICS - Industrial Controls Systems

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 2011 L Street N.W., Washington, DC 20037).

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

*NFPA-70-National Electric Code

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Ave., Boston, MA 02210).

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1 and 6.3), the supplier shall furnish one complete induction heater of the type and size specified in the contract or order for first article inspection in accordance with Section 4.

* 3.2 Design. The induction heaters shall be of the manufacturer's standard commercial product and shall include any additional features necessary to comply with the requirements specified herein. Additional features which are not specified herein that are a part of the manufacturer's standard product shall be included in the equipment furnished. A standard commercial product is a product which has been on the commercial market through advertisements, manufacturer's catalogs or brochures, and represents the manufacturer's latest production models. The heaters shall be designed to convert and control the three phase low frequency (60 Hz) input power to single phase high frequency (3 KHz for Type I and 10 KHz for Type II) output power for the high frequency induction heating work coil at the heating station. The design shall be such as to provide convenient, efficient, and safe operation. All parts subject to wear shall be accessible for adjustment, replacement and repair.

* 3.2.1 Electrical power requirements. The sets shall perform in accordance with the requirements specified herein with an input voltage, and frequency within five per cent of rated value. The MG and starter shall be designed to operate on standard 3 phase, 60 Hz power with input voltages as specified (see 6.2.1).

* 3.2.2 Safety devices. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Safety devices shall not impair the operating functions of the equipment. Electrical interlock safety switches shall be provided for all access doors, the opening of which would expose personnel to voltages in excess of 150 volts. Warning labels shall be attached in conspicuous locations to indicate hazardous conditions. Capacitors larger than 0.1 microfarad

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on circuits above 230 volts shall be provided with means of rapidly draining the stored charge. In addition to other safety requirements specified herein, the induction heaters shall conform to the applicable requirements of NFPA Standard No. 70. The heaters shall comply with the standards promulgated under OSHA applicable to the heater itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

* 3.2.3 Protective devices. Unless otherwise specified (see 6.2.1), protective devices specified in 3.2.3.1 through 3.2.3.3 shall be furnished and installed on each MG set. In addition, where applicable, all protective devices shall terminate into a protective control point-or unit. The control point or unit shall include provisions for determining that protective devices are operative, determining protective devices in operation, and shall include manual reset devices such as switches or push buttons to prevent automatic restarting.

3.2.3.1 Devices for all sizes of MG sets.

a. A vibration eliminator base is to isolate the MG set from the floor or its mounting plate.

b. A bearing electrical insulator to prevent circulating shaft currents during operation.

c. An air temperature switch to stop operation in event of excessive internal air temperatures.

d. A water pressure switch on water cooled units to stop operation in event of inadequate water supply.

3.2.3.2 Devices for control stations.

a. Fuses to protect the control, high frequency meter, and field supply circuits.

3.2.3.3 Devices for heating stations.

a. A water pressure switch to stop operation in the event of inadequate water supply.

b. A water pressure gage to indicate water pressure in the cooling water circuits.

c. A strainer for inlet water to prevent entry of foreign matter into the cooling water circuits. Locate in an accessible position.

d. Sight drains to provide evidence of cooling water being properly provided to critical components.

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3.2.4 Special protective devices. When specified (see 6.2.1), a special rotor support device shall be provided to lock the rotor and to remove the weight of the entire rotor from the thrust bearings. This device is to be used to prevent the brinelling of the thrust bearings that might occur during shipment.

3.2.5 Lubrication. Design of lubrication system shall assure an adequate supply of lubricant to all necessary surfaces and, on all oil lubricated sets, shall include means such as a sight glass for determining the need for replenishment of the lubricant. Means shall be provided to prevent lubricants from entering the electrical system. The lubricant reservoirs shall be of such capacity to provide for minimimum maintenance. A lubrication chart or plate shall be attached to the motor generator set and show the points of lubrication and types of lubricant required.

3.2.6 Cooling systems. Unless otherwise specified (see 3.7.2), the induction heaters shall be primarily water cooled. The complete cooling system shall be non-ferrous, designed to withstand 100 PSIG pressure, and shall operate satisfactorily with water supplied at not less than 30 PSIG pressure and not more than 95°F temperature. A fan located on the main shaft of the MG set shall circulate air through the set and through the finned air-to-water heat exchanger to dissipate the internal heat of the set. Design shall include provisions for water supply inlet at the lowest point to prevent air locks. Components of the heating stations shall be water cooled as required to maintain satisfactory operating conditions and, where applicable, sight drains shall be utilized to ascertain adequate water supply to critical components such as capacitors. When specified (see 6.2.1), a closed water cooling system shall be provided. The system shall be of the type and capacity specified.

3.2.7 Enclosures. All major components shall be enclosed. Enclosures shall provide the necessary protection and support for all components operating at all loads to meet the requirements herein specified. Enclosures shall include means to facilitate ease of handling such as lifting eyes or base arrangement for fork lift use, and ease of maintenance such as access panels and lifting eyes. Unless otherwise indicated, the MG set, control station, and heating station shall be floor mounted.

3.2.8 Dimensions and weight. The units shall be as compact and lightweight as possible consistent with good design and commercial practice. Unless otherwise specified (see 6.2.1), dimensions and weights of units consistent with commercial practice shall be acceptable.

3.2.9 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or of the induction heaters.

* 3.2.10 Materials. Materials not specified herein shall be selected by the manufacturer to meet the requirements of this specification. Contact between dissimilar metals, especially brass, copper or steel in contact with aluminum

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or magnesium shall be avoided whenever possible. Materials shall be free from all defects and imperfections that will affect the serviceability and appearance of the finished product.

* 3.2.11 Reclaimed materials. The heaters may contain reclaimed materials provided such materials will not jeopardize the heaters intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore the material to the same chemical composition and physical properties as the materials that would originally have been selected for use on the heaters.

* 3.2.12 Measurement systems. Unless otherwise specified, either the U. S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U.S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ASTM-380.

3.2.13 Parallel operati (n. MG sets of the same kilowatt (KW) rating and frequency shall be capable of being electrically coupled together to approximately double the KW output. When specified (see 6.2.1), means shall be provided for synchronizing generators when parallel operation is required.

3.3 Construction. The induction heaters shall be constructed so that when installed and connected to the specified power source, they can be used for any operation specified herein. The heaters shall be constructed of parts which are sound in structure and free from any repair.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. In no event shall such processes as welding, peening, plugging, or filling with solders or pastes be used on castings or forgings to reclaim any defective parts for use on the heaters.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where these operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

* 3.3.3 Fastening devices. All screws, pins, bolts, and similar fasteners shall be of a type that resist loss of tightness or installed with means for preventing loss of tightness. Such fasteners subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

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3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the heaters shall be painted in accordance with the manufacturer's standard commercial practice. The interior surfaces of the control and heating station enclosures shall be painted white. If the MG set is enclosed in a sheet metal cabinet, the interior surfaces of the cabinet shall be painted white.

* 3.3.6 Threads. All threaded parts shall conform to FED-STD-H28.

* 3.3.7 Dials. All dials shall be graduated in the inch system. Dial diameters shall be such that the graduations may be easily read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a non-glared background.

* 3.3.8 Plates. All words on instruction and indication plates shall be in the En-language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

* 3.4 Performance. When installed and operated in accordance with the manufacturer's instructions, the high frequency induction heating equipment specified herein shall be capable of converting 60 Hz, three phase input electrical power to nominal 3 KHz or 10 KHz, as applicable, single phase output electrical power for induction heating purposes at the work coil with an overall electrical efficiency of not less than sixty-five per cent. The induction equipment shall be capable of operating continuously as rated power output. Controls shall provide for the equipment to operate satisfactorily from the no-load capacity position to the full-load capacity position. Meters shall indicate all conditions necessary to provide for maximum efficiency and operating conditions. Capacitors shall provide an adequate number of multiple taps to provide for power factor correction. The MG set shall operate with minimum vibration, shall have a frequency output at synchronous RPM of plus or minus four per cent of rated frequency, and motor slip shall not exceed four per cent.

3.5 Components. Unless otherwise specified (see 6.2.1), the heaters shall consist of a motor starter, a MG set, a control station, a heating station, and a high frequency contactor. All electrical components including starters, contractors, relays, switches, push button stations and wiring shall conform to NEMA ICS. Motors and motor-generators shall conform to the requirements of NEMA MG-1,

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3.5.1 Motor starter. Unless otherwise specified (see 3.7.2 and 6.2.1), the motor starters shall be of the full voltage, across the line magnetic type with overload protection, reset button, and a start and stop push button control. Enclosure shall be of general purpose NEMA Type I. The starters shall have a rating suitable for use with the MG set. On some sizes, as an optional feature, the MG start-stop push button station may be included as part of control station (see 3.7.2).

* 3.5.2 MG set. Unless otherwise specified (see 6.2.1), the MG set may be either a vertical or horizontal type and shall be totally enclosed and water cooled with motor and generator rotor on the same shaft. The motors shall be a three phase, 60 Hz, squirrel cage induction type; and the generators shall be a single phase, 3 KHz or 10 KHz as specified. The motor Insulation shall be NEMA Class B, F, or H, and the generator insulation shall be NEMA Class H. Bearings shall be grease or oil type, ball or roller construction, and selected in accordance with bearing manufacturer's recommendations covering application, speed, loading, and minimum life expectancy of 30,000 hours. To minimize the danger of the generator rotor rubbing the generator stator, design of the MG set shall include provisions for the air gap clearance between rotor and stator to increase as operating temperatures increase, and for an air gap clearance of .010 to .035 inch at full load operating temperatures. Design of the MG set shall include ease of maintenance of all components as well as features to minimize damage to rotor or the stator due to bearing failure. The MG shall be vibration isolated from its base or from the floor. Lifting eyes shall be furnished and installed for transporting the MG set or for disassembly of same. The entire shaft assembly including generator rotor, motor rotor, and integral fan shall be dynamically balanced.

3.5.3 Control station. Unless otherwise specified (see 6.2.1), the control station shall be complete with a steel enclosure, the thickness of which shall not be less than 14 gage steel. The enclosure shall be NEMA Type 12 and shall include the following generator controls and monitoring instruments: ammeter, voltmeter, watt meter, and power factor (KVAR) meter calibrated in percentages; and a generator field excitation supply, including rectifier and field relay. Heat "on-off" control with indicating lights shall be included. Output control through generator field shall be provided by continuously variable autotransformer. A control transformer to convert line voltage for control and field supply; indicating lights to indicate control power, station ready, cooling water on, field on, and overload shall be included in the control station. All meters and controls shall be centrally located on the front panel of the

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station in an area convenient to the operator. A high frequency contactor with heat ON-OFF oil tight push buttons shall be furnished separately, as part of control station or heating station, as indicated in 3.5.5.

3.5.4 Heating station. Unless otherwise specified (see 6.2.1), the heating station shall be complete with a NEMA Type I enclosure made of the same material and thicknesses specified for the control station in 3.5.3. The heating station shall be equipped with a power factor correcting water-cooled capacitor with multiple tap arrangement with knife blade switches, jacks, or similar devices for changing taps and output connection to apply maximum generator power to the work coil. The number of capacitors in the heating station will vary depending upon the requirements of the heating application; however, in each heating station there shall be one or more capacitors with at least five tappings to provide for power factor corrections. For optional features available to meet specific operational requirements, see 3.7.2.

3.5.5 High frequency (HF) contactor. A contactor suitable for handling high frequency power shall be furnished to disconnect the HF power supply to the heating station. The HF contactor shall be supplied either in the control station, the heating station, or if necessary, in a separate enclosure. Enclosure shall be NEMA Type I with full length access door with safety interlock switch. When more than one heating station is required, the main HF contactor shall be mounted in a separate enclosure and the line HF contactor shall be mounted in each heating station. Line contractors shall be applicable NEMA Type.

3.5.6 Indicating lights. Unless otherwise specified (see 6.2.1), or indicated herein, indicating lights shall be furnished in accordance with the manufacturer's commercial practice. Indicating lights shall be centrally located convenient to the operator.

* 3.6 Sizes and characteristics. Unless otherwise specified (see 6.2.1), the capacity (KW output) and high frequency output voltages continuously available for each size of induction heater shall be as shown in Table I. The required rated capacities apply to both water and air cooled MG sets.

3.7 Equipment.

3.7.1 Standard equipment. Unless otherwise specified (see 6.2.1), the induction heaters shall be furnished with all standard equipment normally furnished by the manufacturer.

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*TABLE I

3 KHz and 10KHz High Frequency Induction Heaters, Characteristics

Size	Rated Capacity at 1.0 PF* (KW)	High Frequency Voltages Available	Motor Generator Set	
			Maximum Full Load Motor Amperages	Accelerating Time 0 to 3600 RPM (Seconds)
7.5	7.5	220 or 220/440	17 to 18	10
15	15	220 or 220/440	32 to 36	10
20	20	220/440	41 to 46	10
30	30	220/440	56 to 62	10
50	50	220/440 or 400/800	96 to 105	10

*PF Power Factor

3.7.2 Optional features. When specified (see 6.2.1), the following optional features shall be furnished and shall meet the applicable requirements specified herein.

a. Air cooled MG set - air cooled units with or without air filters.

b. Electrical starters - other types of starters required to satisfy local power supply conditions.

c. Control stations with:

(1) MG start-stop push buttons and indicating lights installed in control station.

(2) Automatic voltage regulator to maintain generator output voltage within plus or minus one and a half per cent.

(3) Generator field ammeter.

d. Combination control and heating station - one enclosure embodying all the components and functions of both stations and their optional features.

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e. Heating stations.

(1) Bench mounted or portable (specify details of portability).

(2) Multiple heating stations of the same or different KW output rating for multiple operations.

(3) Push button or foot switch with safety guard, nine foot long three wire rubber covered cable, and twist lock plug with receptacle for manual control.

* (4) High frequency output matching transformer.

(a) Fixed ratio isolating type, water cooled.

(b) Variable ratio isolating type, water cooled.

(c) Variable ratio auto-transformer type, water cooled.

(5) Tap switch for use with (4) (c) above.

(6) Process timer (specify details of timing requirement).

(7) Meters installed on heating station.

(8) Quench system.

(9) Solenoid valve to allow water to flow through work coil when heat is on, with protection to ensure the presence of cooling water flow before heat can be applied.

(10) Heat "ON-OFF" switch and indicating lights.

f. MG set, control station and heating station on common base.

g. Hours of operation meter installed on MG set.

h. Other options as specified.

* 3.8 Suppression of radio interference. When required, radio interference suppression shall be as specified (see 6.2.1).

* 3.9 Moisture and fungus resistant treatment. When required, moisture and fungus resistant treatment shall be as specified (see 6.2.1).

3.10 Portability. Unless otherwise specified (see 6.2.1), portability shall be as indicated in 3.2.8 and 3.5.2.

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* 3.11 Nameplate. A corrosion resistant metal nameplate shall be securely attached to each heater. The nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number
Input power characteristics and rating
Input power factor
Output power characteristics and rating
Output power factor
Date of manufacture
Contract or order number
National stock number or plant equipment code

* 3.12 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.13 Workmanship. Workmanship of the induction heaters and accessories shall be commensurate with the requirements specified herein.

4, QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the first article sample when submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the first article to pass the examination or one or more preproduction tests shall be cause for rejection.

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4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the tests in 4.6.1, 4.6.2, 4.6.3, and the inspection in 4.7. Each item shall comply with all requirements.

4.5 Examination. The induction heating equipment shall be visually examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests. The test requirements indicated below shall supplement but not supersede or replace the supplier's standard equipment test procedures.

4.6.1 Control station. The control and control circuits, safety devices, and protective devices shall be energized to demonstrate compliance with requirements but no actual performance shall be required.

4.6.2 Heating station. The above test shall be duplicated for control station.

4.6.3 optional features. Optional features shall be tested to determine compliance with specified requirements.

* 4.6.4 Performance test. The MG set shall be operated at least four hours. Recorded test data shall indicate compliance with requirements specified herein. The minimum recorded data shall indicate:

a. Resistance of motor, generator, and field windings - hot and cold.

b. Water pressure and quantity, inlet and outlet temperature.

c. Operation of all protective devices.

d. Visual examination for oil or water leaks.

e. Bearing temperatures, air temperatures.

f. Vibration limits taken at top, middle, and bottom - 5 to 15 minutes after start; with up to 2.0 roils (.002 in) being acceptable; 16 minutes after start, up to 1.5 roils (.0015 in) is acceptable.

g. Open circuit and short circuit test.

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h. Heat run, load and temperature test - four hour test run may be extended until consecutive readings one half hour apart indicate stable temperature.

i. High potential test.

j. Motor input - volts, amps, kilowatts.

k. Generator output - volts, amps, kilowatts, and vars.

l. Generator field input - volts, amps.

m. Air gap c earances.

n. Accelerat ng time in seconds.

The overall efficiency, Insulation test, cooling systems, endurance tests, and vibration tests requirements shall be performed and the load and temperature test shall be varied to demonstrate the ability of the multiple tapped capacitor to provide for power factor correction. The output frequency shall be measured.

* 4.6.5 Suppression of radio interference test. Induction heaters having radio interference suppression shall be tested for compliance with 3.8.

* 4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall conform to the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. The induction heating equipment described herein is to be used to produce localized heating of various degrees of intensity for hardening, annealing, brazing, or soldering of metals or some related application of the above.

6.2 Orderin data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

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6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Size, type, capacity, and output voltage required (see 1.2 and 3.6).

c. When first article is required (see 3.1).

d. Motor electrical Input power required (see 3.2.1).

e. Additional safety and health requirements (see 3.2.2).

f. Protective devices, if different (see 3.2.2].

g. Special protective devices, if required (see 3.2.4).

h. Closed water cooling system, if required (see 3.2.6).

i. Dimensions and weights, if special requirements (see 3.2.8).

j. Measurement system, if different (see 3.2.12).

k. Synchronizing equipment, when required (see 3.2.13).

l. Painting, If different (see 3.3.5).

m. Components required, if different (see 3.5).

n. Electric motor starter, if different (see 3.5.1 and 3.7.2).

o. Horizontal or vertical type, if required specify the type (see 3.5.2).

P. Control station, if different (see 3.5.3 and 3.7.3); specify size, capacity, and output voltage.

q. Heating station, if different (see 3.5.4 and 3.7.2); specify size, capacity, and output voltage.

r. Indicating lights, if different (see 3.5.6).

s. Size, capacity, and output voltage required, if other than specified in Table I (see 3.6).

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t. Standard equipment, if different (see 3.7.1).

u. Optional equipment, if required (see 3.7.2).

V. Suppression of radio interference, if required (see 3.8).

w. Moisture and fungus treatment, if required specify (see 3.9).

x. Portability, if different (see 3.10).

y. First article inspection, if different (see 4.3).

z. Quality conformance inspection, if different (see 4.4).

aa. Preparation for delivery, if different (see 5.1).

* 6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

* 6.3 First article. When a first article is required, it shall be tested and approved under the appropriated provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test and approval of the first article.

* 6.4 Safety and health requirements. Paragraph 3.2.3 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electro-magnetic emissions, noxious vapors, air contaminants, in the environment. (Does not limit the hazard level of individual machine in the environment). The procuring agency is advised to analyze the existing hazards levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, the procuring agency should specify the exact configuration of the guard required. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1.e.

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes, additions, modifications, corrections, deletions from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements in this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Review Activities:
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Air Force - 99
Army - AL
DLA - GS

User Activities:
Navy - MC
Air Force - 99
Army - AL

Preparing Activity:
DLA-IP

Project Number:
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